

Isogeny-stratified ECDLP ML blueprint: lead-by-lead triage against the evidence already on file

Date: 2026-10-10. **Kind:** triage of a supplied research blueprint plus two re-analyses of frozen data. **No new measurement was run.** No research state changes here; nothing is approved, promoted or closed by this note. **Claim tier:** toy. **ECDLP claim:** none, in either direction.

Input. "Isogeny-Stratified ECDLP Machine Learning Research, end-to-end technical blueprint, version 1.0, October 2026" (9 pages; SHA-256 276fab868c001bf7d8932a4e64ada7a5e585160c5ea1d6d3cc427c679ffb725). The document calls itself "an implementation-ready specification, not a measured experimental dataset or a demonstration of any cryptographic vulnerability"; its example records are synthetic. It is not committed here (the repository's report pipeline publishes every research/**/*.pdf); the hash identifies it.

Repositories read, at the commits below. Every pointer in this note is a path in one of them.

repo	commit	role
aburan28/crypto	3da0029607dacb6f82aec6ef7f23ad2b7ec1d4d2	Rust cryptanalysis suite, binary and prime isogeny tooling, frozen experiment rows
aburan28/cryptanalysis	605ed9c6b790f078158510fb0bbfca6b240ed0fa	ECC2K-130 class study, C core, mirrored suite
aburan28/ml-cryptanalysis	fe306baf9f7e561f73fdbcb04708f305d896ac726	prime-field curve x invariants x measured-cost program ("CSD")
aburan28/crypto-autoresearcher	85e1d32b7851738cf371a209c6591ba89928df81	ledger, knowledge corpus, frontier map

0. Verdict in one paragraph

The blueprint's leads are not new to this program: each of its four hypotheses has been tested, most of them several times, across the three code repositories and the ledger, and every measured result is a null within the tested scope. The blueprint adds a statistical and ML framing (class-grouped splits, shuffled-conductor nulls, tabular then graph models, active selection). Applying exactly that framing to the frozen rows that already exist gives the same answer: on the only binary isogeny class on file with two floor levels, volcano position explains 0.14 % of the variance in relation rate beyond factor-base size (permutation $p = 0.36$); on 1,548 prime-field curves with conductor, class-group and volcano features, the feature block changes out-of-fold R^2 by 2×10^{-5} against controls that already reach 0.997. The blueprint's own stop rule ("No verified relation advantage: publish null result and stop expensive scaling", §19) therefore applies before its phases 4 to 6 (tabular ML, graph ML, productisation) are built. What is genuinely open is narrower than the blueprint and is stated in §4, with the records that already carry it.

1. The four hypotheses against the evidence

Labels: **measured** = a run with retained artifacts; **derived** = a proof or exact computation; **independently verified** = a second implementation or reviewer re-derived it; **proposed** = a design that has not run.

blueprint hypothesis	status	strongest evidence (path, scope)
H0 conductor and graph features have no incremental predictive value beyond matched controls	supported in every tested scope (prime-field rho; binary summation-polynomial index calculus at $m = 2$; $n = 131$ class structure)	§3 of this note (measured, re-analysis); ml-cryptanalysis/experiments/anomaly-search/report.md (measured: 1,545 curves, 0 replicated anomalies, neighbour cost ratios within error of 1 after the Aut discount); ml-cryptanalysis/experiments/isogeny-structure/report.md Stage B (measured: best cost-residual formula over 57 invariants at train R^2 0.015 against a permutation-null 95th percentile of 0.017, 0 formulas above the null band); crypto/research/notes/koblitz-isogeny/presentation-vs-curve-results-20261004.md (measured: curve intraclass correlation of index-calculus cost ≤ 0.09 on every valid metric at $n = 16$ to 23)

blueprint hypothesis	status	strongest evidence (path, scope)
H1 conductor valuation and topology predict relation yield, degree of regularity or total solver cost on unseen classes	rejected in scope for the degree metrics (derived: theorem) and for yield (measured: explained by a setup quantity)	Degree of regularity and first-fall degree: the leading forms of the Weil-descended Semaev systems on $y^2 + xy = x^3 + ax^2 + b$ are free of b for every m (Boundary C', proved by induction in <code>crypto/research/notes/ecc2k130/RESEARCH_ISOGENY_CLASS_SEARCH.md §2C'</code> , EXP-R6d), and the Macaulay rank profile at degrees 2 to 4 is identical on every member and twist of the n = 17, 19 classes walked (<code>crypto/RESEARCH_PDP_SPEEDUP_AND_ISOGENY.md §2.3</code> , measured). Yield: the n = 31 spread (variance ratio 4.1 over 100 class members) is reproduced with no fitted constant by the cofactor balance of the fixed factor base, hold-out correlation 0.850 against 0.342 for the $ F ^2$ null (<code>crypto/research/notes/koblitz-isogeny/yield-spread-results-20261003.md</code> , measured on a pre-registered hold-out). Prime fields: H-ISO-001 rejected_scoped (ledger/EV-ISO-001.yaml, toy p $\approx 2^{13}$, l $\in \{2, 3, 5\}$, m = 3: d_reg = 2 on every neighbour and control, neighbour yields inside the control band)
H2 effects replicate across binary Koblitz, general binary ordinary, prime-field and prime-field-extension curves	null replicates on three of the four families; the fourth has no isogeny-stratified measurement and no mechanism	Koblitz classes n = 16 to 31 and the ECC2K-130 class at n = 131 (<code>cryptanalysis/experiments/ecc2k130-isogeny-class/data/per_curve_hardness.csv</code> , 263 level-labelled curves; <code>cryptanalysis/README.md</code> status note: E0 needs 0.996 x the attempts of its descendants, 95 % CI 0.976 to 1.017, n = 19 end to end); random-b binary ordinary classes at n = 17, 19 (<code>crypto/RESEARCH_PDP_SPEEDUP_AND_ISOGENY.md</code>); prime fields (ml-cryptanalysis, EV-ISO-001, EV-VOLC-3529e8). Prime-field extensions $F_{\{p^n\}}$: GOAL-GFPN-380702 has no isogeny-stratified cell, and the only model-dependent methods there (covers, Weil descent) are already the subject of IDEA-20260920-10547b and the frontier rows KR-IC-5ea2f8, KR-IC-791cb7
H3 direct attack advantages remain interesting after pricing isogeny construction and subgroup transfer	moot where measured: there is no direct advantage to price; transfer is priced anyway	n = 131: every constructible class member reaches E0 by one ascending 263-isogeny at $\leq 2^{15.6} F_q$ -multiplications, effective hardness 60.81 bits on all 263 curves (<code>cryptanalysis/experiments/ecc2k130-isogeny-class/data/hardness_by_level.csv</code> , derived and independently re-derived); the two large components separated by a 57-bit prime conductor gap cannot be written down (<code>crypto/docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md §1</code> , derived). Transfer-cost pricing for large prime degrees is pre-registered as protocol I-2 in <code>crypto/research/isogeny_class_difficulty_20261008/</code> (proposed, unrun)

The theoretical frame the blueprint is missing is in the repositories: within an F_q -isogeny class, ECDLP difficulty is invariant up to transport cost (`crypto/research/isogeny_class_difficulty_20261008/README.md`, "The theorem that frames every experiment"; frontier row KR-RHO-755e35; EV-IT-511f3d, strong, closing the isogeny-transfer lane by Tate's theorem). Only three things can vary inside a class: transport cost, the automorphism group ($j \in \{0, 1728\}$), and model-dependent non-generic structure that a method reads off the equation rather than the group. Conductor strata can affect attack cost only through the third channel, and for summation-polynomial index calculus that channel is the constant term b, which Boundary C' shows does not reach the leading forms. The one place the literature records class-internal variation in attack cost is GHS Weil descent over composite binary degree, where the magic number is model-dependent and horizontal walks find the weak members; that is a frontier row (KR-IC-791cb7), not an open lead.

2. Lead-by-lead map of the blueprint's sections

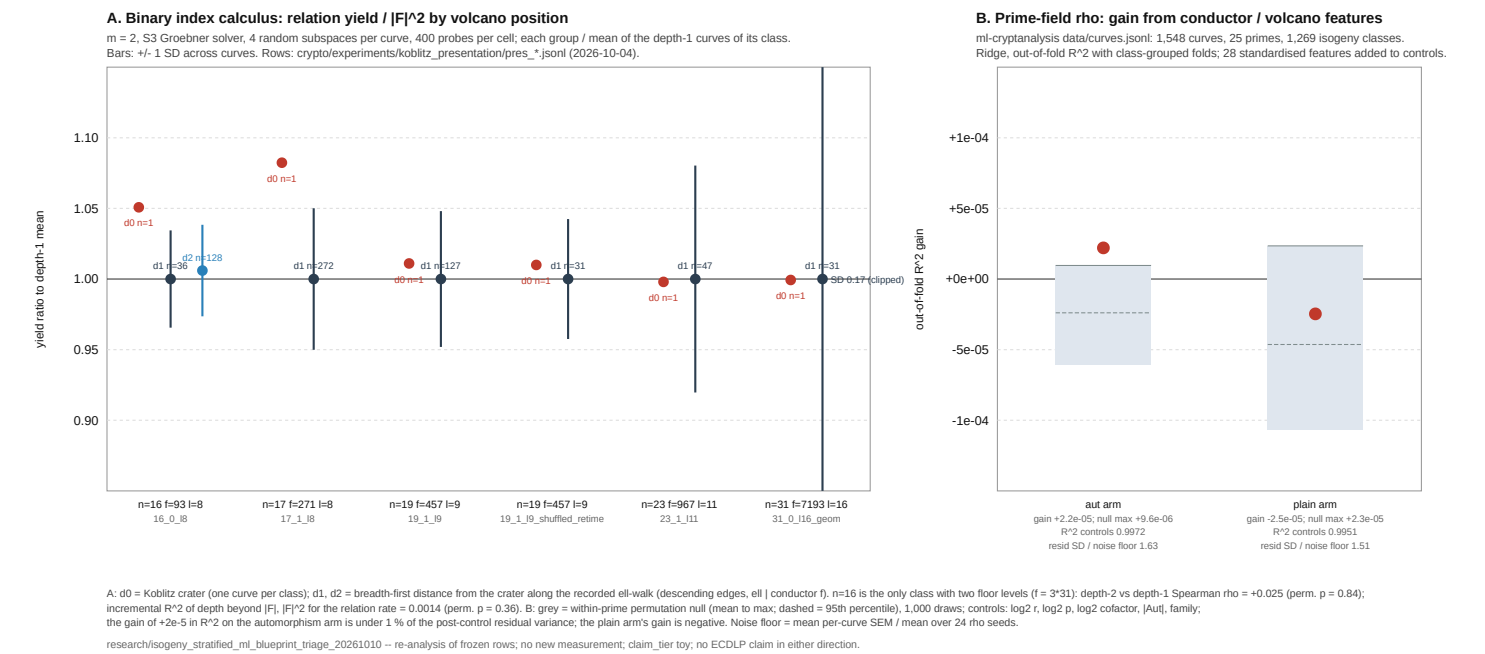
blueprint section	what already exists	gap
§2 arithmetic certification (order, ordinarity, D_K , f_π , f_E , edge certification, $\gcd(\text{degree}, r) = 1$)	<code>crypto/src/cryptanalysis/curve_traits/</code> ($t^2 - 4q = v^2 d_K$ split, conductor, per-l depth and splitting, End bounds, tests); <code>crypto/isogeny_algos/src/path/endo.rs</code> (Kohele's End(E), tested); <code>ml-cryptanalysis/csd/endo.py</code> (exact f_E , verified against Deuring's class-number count); <code>crypto/src/cryptanalysis/isogeny_walk/</code> (every edge kernel-certified, subgroup transport checked; 325,000 certified edges over P-256/P-224/P-192 walks); <code>cryptanalysis/experiments/ecc2k130-isogeny-class/ground_truth/</code> (262 floor curves at n = 131, j-invariants agreeing between PARI polclass and Vélú)	none of substance; Fouquet–Morain is cited but not separately implemented (Kohele's method and the Sutherland-style walks cover the use)
§3, §6, §7 entities, IDs, JSON contracts, SQL schema	<code>crypto/docs/curves/ICV1.md + registry.json</code> (321 curves), <code>traits.js</code> on with <code>frobenius{disc, cm_disc, conductor, ...}</code> and <code>small_primes[{ell, depth, splitting}]</code> , <code>docs/curves/ic/curves.yaml + curves.schema.json</code> (EC1 identities), <code>docs/ecbench/schema.sql</code> (curves, representations, workloads, factor bases, sessions, arms, runs, phases, claims; views <code>ic_phase_split</code> , <code>ic_yield</code>), exact integers as strings, SHA-256 content hashes, IW1 route records	the blueprint's PostgreSQL/S3/Redis service layer does not exist and is not needed by any measurement on file

blueprint section	what already exists	gap
§4, §5 feature dictionaries	static: <code>ml-cryptanalysis/csd/invariants.py</code> (57-invariant vocabulary incl. <code>vol_level/vol_height</code> per ℓ , class-group exponent, 2-rank, <code>ord_\ell</code> , torsion, embedding degree, twist order) and <code>crypto/docs/curves/TRAITS.md</code> ; experimental: <code>ecbench</code> phase records, <code>koblitz_isogeny_cost</code> rows (unknowns, relations, independent/dependent/inconsistent, Gröbner ns and calls, reductions, first-fall histogram, D^* histogram, BSGS cross-check)	descent namespace (tower degrees, Weil-descent dimension, cover genus) is computed ad hoc in <code>ghs_screen.rs</code> and the GFPN notes, not as a per-curve table
§8 data generation (classes, curves, graph, publish)	binary: exhaustive trace-scan class census and $\Phi_\ell \bmod 2$ walk (<code>crypto/src/cryptanalysis/isogeny_class_search.rs</code> , <code>binary_isogeny.rs</code> , <code>experiments/koblitz_isogeny_class_walk.json</code> : 7 full classes with conductor, ℓ s, edge lists); prime: <code>isogeny_walk</code> (deterministic run ids, <code>verify</code> replay); $n = 131$: <code>v2_descend_toy.py</code> smoke descent PASS at $n = 23$ (E0 in <code>crypto/docs/ic/EXPERIMENTS_ISOGENY_CONDUCTOR_GAP_20261007.md</code>)	the blueprint's binary targets $m \in \{31, 51, 53, 83\}$: 31 and 53 and 83 have classes and rungs; $m = 51$ has nothing ; multi-level binary classes with constructed curves exist only at $n = 16$ ($f = 3 \cdot 31$) in-repo and as the off-repo class C37 (four levels, <code>ic-conductor-leads/data/class37.json</code> , not in these checkouts)
§9 paired benchmarking (immutable manifests, interleaved A/B, pinned CPU, exact accounting)	<code>ecbench</code> (<code>crypto/docs/ecbench/</code> , skills <code>ecbench-measure</code> , <code>ecbench-bounds</code> , <code>ecbench-independent-runner</code>): sealed sessions, replay certificates, paired bootstrap intervals, counted group operations as the unit, isolation grades; <code>PLAN_IC_ACCOUNTING_FIXES_20261007.md</code> (≥ 30 targets, equal precompute, operation counts beside wall clock)	none; <code>ecbench</code> exceeds the blueprint's protocol
§10 targets and cost formulas (T_{total} decomposition, <code>pair_log2_cost</code> , censoring)	<code>ecbench</code> phases and $S = GAE/\sqrt{r}$; censoring handled as explicit timeout rows (<code>timeout_flag</code> , failure reason) in <code>koblitz_isogeny_cost</code> and the ledger's run manifests	none
§11 splits, leakage, uncertainty	<code>ml-cryptanalysis/csd/splits.py</code> (isogeny-class and field-family splits, label-permutation null), hidden-vs-visible vocabularies, fresh-holdout verification (<code>csd/verify.py</code>); presentation study's crossed random-effects model with 2,000 curve bootstraps	none
§12 ML stages (control $\rightarrow$ tabular $\rightarrow$ GNN $\rightarrow$ multi-task $\rightarrow$ active)	control and tabular done: ridge and MLP baselines, symbolic predicate search with permutation bands (<code>ml-cryptanalysis</code>); EXP-RELN-82f487 compared a message-passing GNN against gradient-boosted trees as a decomposability predictor (EV-RELN-cc045d, EV-RELN-99c688: inconclusive)	GNN over the isogeny graph and active selection were never built, because the control and tabular stages found nothing to learn (this note §3 confirms)
§13 active experiment selection	not built	dominated: the information gain of a certified pair is bounded by a within-class variance that is at the noise floor (§3)
§14, §15 repository and service architecture, API, state machine	the three repositories already split "certified math" (<code>crypto</code>) from orchestration and records (<code>crypto-autoresearcher</code>), with the ledger's state machine (<code>AGENTS.md</code> , "Research states")	the HTTP API, queue and UI do not exist and no measurement needs them
§16 validation matrix	arithmetic, edge, serialization and reproducibility checks: <code>isogeny_walk_verify</code> , <code>ecbench_verify --replay-all</code> , ICV1 identities, <code>tools/validate_ledger.py</code> ; statistical controls: label permutation, same-stratum pairs, isomorphic duplicates ($n = 16$ $a_2 = 1$ run)	operational (worker interruption, duplicate submission) only inside taskq, untested live
§17 phases 0 to 6	phases 0 to 3 effectively exist across the repos; phase 4 (tabular) exists for prime-field ρ	phases 5 and 6 blocked by the phase-4 null, per the blueprint's own §19 rule
§18 pilot datasets (≥ 10 certified classes per family, 3+ seeded trials per pair)	binary: 7 full classes ($n = 16$ to 23) plus sampled $n = 31$ and the $n = 131$ class; prime: 1,277 isogeny classes over 25 primes with 24 ρ seeds per curve	"no conductor variation" (§19 risk 1) is the binding constraint: Koblitz classes have one crater over one floor except $n = 16$
§19 risks and stop rules	the measured record triggers rule 5 ("no verified relation advantage")	—
§20 definition of done	items 1 to 4 and 6 are met by existing artifacts; item 5 (UI) is not and is not warranted	—

blueprint section	what already exists	gap
§21 literature checklist	corpus entries: Kohel inside KN-LIT-319 (no standalone entry), Sutherland KN-LIT-360, KN-LIT-251, KN-LIT-284, Galbraith KN-LIT-7630, KN-LIT-ca35e0, KN-LIT-3748, GHS KN-LIT-007, Gaudry KN-LIT-002, Diem KN-LIT-003, Faugère KN-LIT-027/028; frontier rows KR-IC-5a43e5, KR-IC-5ea2f8, KR-IC-791cb7, KR-IC-f5c584, KR-RHO-755e35	Fouquet–Morain has no dedicated KN-LIT entry (cited in KN-LIT-319 and KN-TECH-018); Kohel's thesis has none

3. Two re-analyses of frozen rows with the blueprint's own controls

Both scripts are in this directory and read only committed data in the sibling repositories; both finish in seconds on four cores. Results are the JSON files beside them; the figure is figures/level_stratified_null.svg.



3.1 Binary index calculus: volcano position against yield, reductions and time

Data. crypto/experiments/koblitz_presentation/pres_*.jsonl, the rows of the presentation-vs-curve study (design 2026-10-03, results 2026-10-04). Each row is one (curve, random F_2 -subspace V) cell: m = 2 Semaev decomposition with the S_3 Gröbner solver, 400 full-group probes, four seeded subspaces per curve, every transported logarithm BSGS-checked. The depth field is the breadth-first distance of the curve from the Koblitz crater along the recorded ℓ -walk edges of experiments/koblitz_isogeny_class_walk.json (examples/koblitz_presentation_sweep.rs); for ℓ | f those edges are descending, so depth is the curve's volcano position. The source study measured the curve-versus-setup variance split and, by its own design, ran the level regression only if a curve effect appeared; none did, so this is the first time the level label is used on these rows.

Method (level_stratified_analysis.py). Average the four cells per curve; compare the crater (one curve per class) to the floor distribution by z-score; among floor curves test depth against each metric by Spearman correlation and by difference of group means, each with a 20,000-draw permutation null that shuffles depth across curves; test the incremental R^2 of depth for the raw relation rate beyond $|F|$ and $|F|^2$. Timing metrics from curve-order runs are flagged as confounded, as the source note established.

Result (full table: level_stratified_results.md).

class	conductor f	walk ℓ	curves by depth	yield/ $ F ^2$: crater z vs floor	depth among floor (yield)	depth beyond $ F $ for relation rate
n = 16, l = 8	93 = 3·31	3, 31	d0: 1, d1: 36, d2: 128	+1.40 (91st pct)	ρ = +0.025, p = 0.84; Δ (d1 – d2) = –0.18 SD, p = 0.33	R^2 0.741 → 0.742, Δ = 0.0014, p = 0.36
n = 17, l = 8	271	271	d0: 1, d1: 272	+1.64 (95th pct)	one floor level	—

class	conductor f	walk ℓ	curves by depth	yield/ F ²: crater z vs floor	depth among floor (yield)	depth beyond F for relation rate
n = 19, l = 9	457	457	d0: 1, d1: 127 (32 re-timed)	+0.23	one floor level	—
n = 23, l = 11	967	967	d0: 1, d1: 47	−0.02	one floor level	—
n = 31, l = 16 (geometric V)	7193	7193	d0: 1, d1: 31	−0.00	one floor level	—

Reductions per call and (where valid) log time per call behave the same way: every crater z lies in [−1.03, +0.61], every floor-depth test at n = 16 has $p \geq 0.33$. The crater's yield sits at the 91st to 95th percentile of its floor at n = 16 and 17 and at the median at n = 19 to 31; across five classes that is consistent with chance and, if anything, points the way the τ -structure argument predicts (the crater is the easiest curve of its class by every known measure, `crypto/docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md` §1 item 4), never toward a weaker descendant.

Reading. Measured, toy, $m = 2$, this solver, these subspaces. Within the scope the blueprint's H1 is not supported for any of its three named targets; depth adds nothing to |F| for yield, and the degree metrics are theorem-constant anyway. The n = 16 class is the only one on file with two floor levels, so the depth-1 vs depth-2 contrast rests on 36 against 128 curves at one field size; it cannot speak to $m \geq 3$ or to $n \geq 31$.

3.2 Prime-field rho: conductor, class-group and volcano features beyond matched controls

Data. `ml-cryptanalysis/data/curves.jsonl`: 1,608 curves over 25 primes (2^{15} to 2^{20}), 1,277 isogeny classes, families random / class-number-one CM / $j = 0$ / $j = 1728$, with exact f_E (certified on 1,545), `vol_level` and `vol_height` for $\ell \in \{2, 3, 5, 7\}$, class-group structure, rational isogeny counts, and counted Pollard-rho cost S (group operations / $\sqrt{r}$) averaged over 24 seeds, both with the automorphism-class walk and the structure-blind plain walk. Supersingular rows and rows without both arms are dropped (1,548 used).

Method (`prime_field_nested_test.py`). Ridge regression of $\log S$ on controls only ($\log_2 r$, $\log_2 p$, $\log_2$ cofactor, |Aut| one-hot, family one-hot) and on controls plus 28 standardised candidate features ($\log_2 f_E$, $\log_2 f_\pi$, $v_2(f)$, $v_3(f)$, End maximal, $\log_2 h_{\text{End}}$, $\log_2 h_K$, rational isogeny counts up/down/horizontal, 2-torsion rank, 3-torsion, class-group 2-rank and cyclicity, `vol_level`, `vol_height` and Kronecker symbol at each ℓ). Out-of-fold R^2 with five folds that keep each isogeny class in one fold. Null: the candidate block permuted across curves within the same prime, 1,000 draws. The per-curve standard error of the mean gives the noise floor.

Result.

target	OOF R^2 controls	+ candidates	gain	null (mean, 95th pct, max)	residual SD after controls	mean SEM/mean	univariate gains
log S, automorphism- class rho	0.9972	0.9972	+2.2 x 10 ^{−5}	−6.1 x 10 ^{−5} , −2.4 x 10 ^{−5} , +0.96 x 10 ^{−5}	0.025	0.015	$\log_2 f_E$, <code>vol_level</code> ₂ , <code>vol_level</code> ₃ , $\log_2 h_{\text{End}}$, <code>n_down</code> : all $ \Delta < 10^{-5}$
log S, plain rho	0.9951	0.9951	−2.5 x 10 ^{−5}	−10.7 x 10 ^{−5} , −4.6 x 10 ^{−5} , +2.3 x 10 ^{−5}	0.031	0.020	same

Reading. Measured, toy. The automorphism arm's gain is formally above the permutation null's maximum, but it is 2×10^{-5} in R^2 , under one percent of the post-control residual variance, and the plain arm's gain is negative; no single conductor or volcano feature carries a univariate gain above 10^{-5} . The residual after controls is 1.5 to 1.6 times the seed-averaging noise floor, so some unmodelled variance remains (cofactor and r -dependence are entered log-linearly), but none of it is captured by the blueprint's feature families. This is the blueprint's H0 with its own leakage firewall (no same-run quantities in the features, class-grouped folds, label permutation), and it agrees with the anomaly search and the Stage B cost null already in that repository.

4. What remains open, and why no new proposal is filed from this note

The ledger already holds 61 proposed ideas under RQ-VOLC-f6253b (the research question that mandates stratifying every measurement by level and by End-ring conductor), plus RQ-ICINV-475b5e (any cost functional non-constant across a class) with three evidence records at neutral or preliminary strength and one refutation (EV-ENDO-eb58c4). Under "Approval is bounded by execution" and "never dispatch a task you cannot rank ahead of doing nothing" (AGENTS.md), adding records that the frontier map already forecloses would be noise. The residue, after the dedup in this session (tools/build_frontier_map.py --match, ledger/.index/proposals.jsonl):

residual lead	why it is still open	dominated by / carried by	cheapest discriminating test
R1. Multi-level binary panel at $m \geq 3$ with the production solvers. The ICC study dropped $m \geq 3$ for budget (≈ 295 s per call at $n = 17$) and the first E1 panel ran $m = 2$ on T23 only.	No row on file measures level against solver cost at the arity where the Gröbner step is non-trivial.	Degree metrics: Boundary C' (theorem) predicts no level effect; yield: $ F ^m$ scaling and cofactor balance predict it. Carried by crypto/docs/ic/EXPERIMENTS_ISOGENY_CONDUCTOR_GAP_20261007.md E1 (T19, C37) and E3, and protocol I-4 in crypto/research/isogeny_class_difficulty_20261008/; no ledger record.	E1 on class C37 (levels 1, 73, 2663, 73-2663) at $m = 3$, one curve per level, ≥ 30 targets, level-shuffled null; expected null; cost 10^2 to 10^3 core-hours. It calibrates the instrument; it cannot produce a hardness claim because transport prices every level back to E0.
R2. Model-dependent screens over a whole class (GHS magic number, cover existence, subfield-definedness) for families where such methods exist: composite-degree binary and $F_{\{p^n\}}$.	The one mechanism by which class position is known to change attack cost; measured densities exist only for binary composite degree.	Known mechanism: KR-IC-791cb7, KR-IC-5ea2f8; carried by IDEA-20260920-10547b (odd-characteristic GHS/Diem), IDEA-20261002-26c628 (cover genus), IDEA-20260922-d11575 / -0e3641 (ANSI binary curves), protocol I-3.	Already specified in those records; nothing in the blueprint changes their ranking.
R3. Prime-field extension leg of H2.	No isogeny-stratified cell exists for $F_{\{p^n\}}$ (GOAL-GFPN-380702).	No mechanism for conductor to enter rho or decomposition cost there beyond R2.	Not worth a cell on its own; folds into R2 if R2 is ever run.
R4. The JMV open case (two curves in one class whose End-ring conductors differ by a large prime; no reduction known either way).	Genuinely unknown; realised exactly by the ECC2K-130 class.	Structurally unobservable at $n = 131$ (neither side can construct a curve across the gap); V6 in RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md; KN-OPEN-cbbd97 (End-ring-gated walks); KN-LIT-ca35e0 (Galbraith 2024, climbing and descending tall volcanoes).	Theory, not measurement: a polylog-in- ℓ representation of a rational prime-degree isogeny from a CM surface when π acts as a scalar on $E[\ell]$. Nothing known.
R5. Blueprint phases 5 and 6 (GNN on the isogeny graph, active selection, service layer).	—	Dominated by §3: there is no signal to learn at the noise floor, and the blueprint's §19 says stop.	None.

The conductor-gap, class-walk and presentation studies in crypto have pre-registered the remaining measurements (E1 to E9, I-1 to I-5) with pass/fail predictions written before any run. If a Coordinator chooses to spend on R1, those protocols are the contracts to freeze; this note adds the level-shuffled null and the class-grouped bootstrap as required controls.

5. Records and paths this note rests on

Ledger: RQ-VOLC-f6253b, RQ-ICINV-475b5e, RQ-ISO-001, RQ-RELN-61638a, RQ-ICPERF-94c86e; GOAL-ENDO-001, GOAL-ECDLP2M-001, GOAL-ICPERF-e6b6a4, GOAL-GFPN-380702; H-ISO-001 (rejected_scoped, DEC-20260716-003), EV-ISO-001, EV-VOLC-3529e8 (strong, neutral), EV-ICINV-343679, EV-ICINV-6b6264, EV-ICINV-8afb64, EV-ICINV-c68f13, EV-ENDO-eb58c4, EV-IT-511f3d (strong, contradicts), EV-RELN-cc045d, EV-RELN-99c688; IDEA-20260807-3f5545, IDEA-20260815-101067, IDEA-20260807-b1b604, IDEA-20260920-10547b, IDEA-20261002-26c628; KN-FIND-f293c6, KN-OPEN-cbbd97, KN-OPEN-7f0d85; frontier rows KR-RHO-755e35, KR-IC-791cb7, KR-IC-5ea2f8, KR-IC-f5c584.

crypto: RESEARCH_PDP_SPEEDUP_AND_ISOGENY.md; docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md; docs/ic/EXPERIMENTS_ISOGENY_CONDUCTOR_GAP_20261007.md; research/notes/ecc2k130/RESEARCH_ISOGENY_CLASS_SEARCH.md; research/notes/koblitz-isogeny/{yield-spread-design,yield-spread-results}-20261003.md, presentation-vs-curve-{design-20261003,results-20261004}.md; research/isogeny_class_difficulty_20261008/README.md; experiments/koblitz-presentation/pres_*.jsonl, experiment

s/koblitz_isogeny_class_walk.json, experiments/koblitz_isogeny_cost_sweep.json; docs/isogeny-walk/README.md; docs/curves/{registry.json, traits.json, ICV1.md}; docs/ecbench/schema.sql; src/cryptanalysis/{koblitz_isogeny_cost, isogeny_classes_search, binary_isogeny, curve_traits}; isogeny_algos/src/path/{endo, volcano}.rs.

cryptanalysis: README.md (status note 2026-10-01); experiments/ecc2k130-isogeny-class/{REPORT.md, data/per_curve_hardness.csv, data/hardness_by_level.csv}.

ml-cryptanalysis: README.md; csd/{invariants, endo, isogeny, splits, rho}.py; data/curves.jsonl; experiments/{anomaly-search, isogeny-structure, planted-aut}/report.md.

6. Files in this directory

file	content
README.md	this report
level_stratified_analysis.py, level_stratified_results.json, level_stratified_results.md	§3.1 script and output
prime_field_nested_test.py, prime_field_nested_results.json	§3.2 script and output
make_figure.py, figures/level_stratified_null.svg	the quantitative graph (editable source is the script)
README.pdf	rendering of this report with the figure